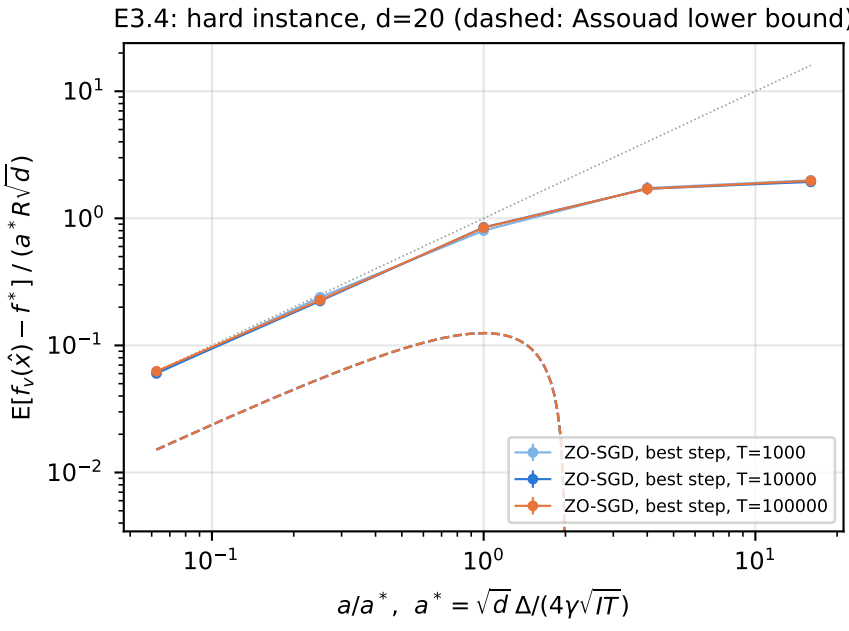
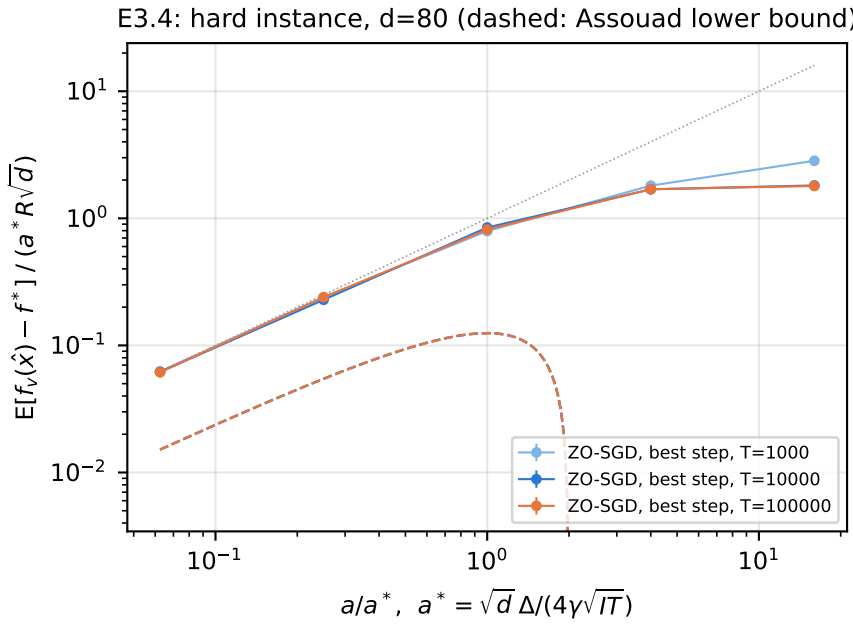

E3.4: hard instance, d=20 (dashed: Assouad lower bound)
10^1
10^0
10^{-1}
10^{-2}
$\mathrm{E}[f_v(\hat{x}) - f^*] / (a^* R\sqrt{d})$
10^{-1}
10^0
10^1
$a/a^*,\ a^* = \sqrt{d}\,\Delta/(4\gamma\sqrt{IT})$
ZO-SGD, best step, T=1000
ZO-SGD, best step, T=10000
ZO-SGD, best step, T=100000


E3.4: hard instance, d=80 (dashed: Assouad lower bound)
10^1
10^0
10^{-1}
10^{-2}
$\mathrm{E}[f_v(\hat{x}) - f^*] / (a^* R\sqrt{d})$
10^{-1}
10^0
10^1
$a/a^*,\ a^* = \sqrt{d}\,\Delta/(4\gamma\sqrt{IT})$
ZO-SGD, best step, T=1000
ZO-SGD, best step, T=10000
ZO-SGD, best step, T=100000